lab 2 Performance

Start Assignment

- Due Thursday by 5pm
- Points 100
- Submitting a file upload

In this lab, you will design a set of benchmarks and measure their performance on the OTTER (our 233 processor). The set of bench- marks (e.g. matrix multiplication and addition) are representative of many important applications that are used in systems today. This lab will give you more practice with assembly programming, the RISCV ISA, and the basic multi-cycle processor design. After implementing and testing your benchmarks, for the second part of the lab you will measure the performance of the multi-cycle processor with a 1-cycle latency memory . Learning Objectives:

- Understand basics of an instruction set architecture
- Understand a basic multicycle processor microarchitecture
- Abstraction levels, including register-transfer-level modeling
- Design principles, including modularity, hierarchy, and encapsulation
- Design patterns, including control/datapath split

How to use Git:

[Guide to using Git for CPE-333](https://youtu.be/CiUSIBvFIIo) (https://youtu.be/CiUSIBvFIIo)

Please consider using GitHub for all .sv source files.

TODO:

PART 1: Compile/Assemble using RARS (you must use the RARS version provided on this course main page contains the memory specific for the OTTER) the Matrix Mult Program you created on lab0.

Note: you need to create the RISC_V machine code file, mem.txt‘ file which is the memory image for use in the simulator.

Load this file into the memory file in your Vivado ->Project ->RISCMultiCycle and simulate the execution of this MM program on your Otter CPU.

Create the waveform for the MM and copy paste part of the waveform showing that is working (you can annotate the image)

Provide the CPU Execution time , Power and Area

PART 2: Setup the baseline OTTER (multi-cycle) and verify the processor with the testall.s program on Canvas. You can use your OTTER design from 233 or the reference design posted on Canvas ($OTTER - multicyle - 1 - cycle - $memory.zip).

  - Measure the performance of the testall program (e.g. to cycle through all instruction test cases, the total number of cycles).
  - Measure the performance of your benchmarks from part 1 on the OTTER, for matrices of size (3x3, 10x10, and 50x50).

Files you <u>may find helpful</u>:

- Matrix Initialized to random integer numbers [datasets.zip](https://canvas.calpoly.edu/courses/186931/files/21847760?wrap=1) (https://canvas.calpoly.edu/courses/186931/files/21847760?wrap=1) (https://canvas.calpoly.edu/courses/186931/files/21847760/download?download_frd=1)
- Matrix Multiplication code in C (with some initialization) [matmul.tgz](https://canvas.calpoly.edu/courses/186931/files/21847772?wrap=1) (https://canvas.calpoly.edu/courses/186931/files/21847772?wrap=1) (https://canvas.calpoly.edu/courses/186931/files/21847772/download?download_frd=1)

- Test all programs for Otter testing on Basys Board(upload a video of test_all running on bysis board) **test_all.zip (https://canvas.calpoly.edu/courses/186931/files/21847854?wrap=1) (https://canvas.calpoly.edu/courses/186931/files/21847854/download?download_frd=1)**
- Staring Verilog template file **simTemplate-1.v (https://canvas.calpoly.edu/courses/186931/files/21847917?wrap=1) (https://canvas.calpoly.edu/courses/186931/files/21847917/download?download_frd=1)**
- Otter Architectural Diagram **Otter_arch_v1.7b.pdf (https://canvas.calpoly.edu/courses/186931/files/21847775?wrap=1) (https://canvas.calpoly.edu/courses/186931/files/21847775/download?download_frd=1)**
- **For VIVADO**: TA's Otter Implementation **OTTER_Diego_Curiel.zip (https://canvas.calpoly.edu/courses/186931/files/21847904?wrap=1) (https://canvas.calpoly.edu/courses/186931/files/21847904/download?download_frd=1) (https://canvas.calpoly.edu/courses/186931/files/21847943?wrap=1)**

UPLOAD:

Report : pdf document with names of students on the group and what each of the members did for the project plus the number of hours spend by each team member

screenshots of the waveforms for the simulations (MatMult and test_all)

execution time for the mat mult and the testall

zip file with all the project source files so it can be replicated

video link or demo TA/Instructor : You project working on the microchip board